MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, LOW NOISE, 50 OHMS, M17/132-RG404

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

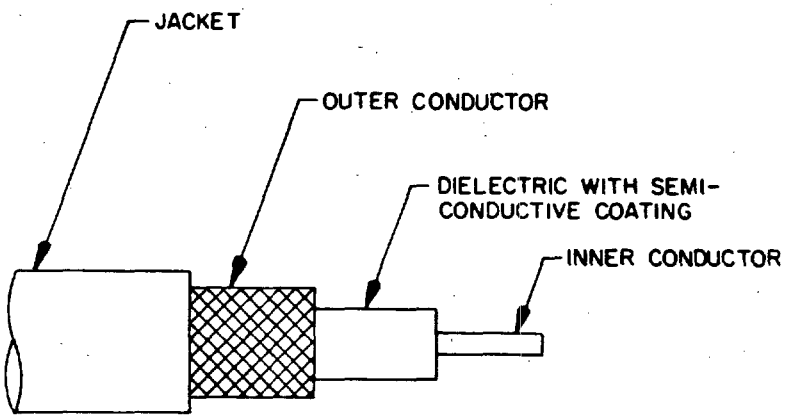


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of AWG size 38 silver-plated, copper-covered steel wire. Overall diameter: 0.012 inch.
Dielectric	Type F-1 with a semiconducting coating. Overall diameter: 0.033 inch ±0.002.
Outer conductor	Silver-coated, AWG size 38 copper strands. Overall diameter: 0.056 inch, maximum. Carriers: 16 Ends: 3 Picks/inch: 25 ±10%
Jacket	Type IX. Overall diameter: 0.071 ±.004 inch.

FSC 6145

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Engineering information:

Continuous working voltage: 1,500 Vrms, maximum.
Operating frequency: 1 GHz, maximum.
Velocity of propagation: 68 percent, nominal.
Operating temperature range: -40° to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 24.45 ohms per 100 feet.
Elongation: 10 percent, minimum.
Tensile strength: 50 $klb_f/inch^2$, minimum.
Engineering notes: This cable useful in low noise temperature applications.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors:
Inner conductor to core: 2 pounds minimum, 8 pounds maximum.
Aging stability: Not applicable.
Stress crack resistance: +230° ±5°C for 76 hours.
Outer conductor integrity: Not applicable.
Cold bend: -55° ±2°C.
Dimensional stability: +200° ±5°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Weight: 1.8 pounds per 100 feet, maximum.

Electrical:
Test frequency: 50 MHz to 1 GHz.
Spark test: 2,000 Vrms +25%, -0%.
Voltage withstanding: 1,500 Vrms.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,000 Vrms, minimum.
Characteristic impedance: 50 ohms ±2.
Attenuation: 29 dB/100 feet maximum at 400 MHz.
Capacitance: 31.5 pF per foot, maximum.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: 25 millivolts, peak to peak maximum.
Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/132-RG404	RG-404/U

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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